

# Sanjana Gautam, PhD

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## Summary

Sanjana Gautam is a Senior Applied Scientist at Microsoft, where she works on the scientific evaluation of the Finance Agent, an AI agent supporting finance professionals across Microsoft 365 and Dynamics 365. Her work designs rigorous methods such as LLM-as-a-judge evaluation, automated experimentation, and systematic prompt and data tuning to measure and improve how AI agents perform in real workflows. Drawing on a background in human-computer interaction, AI ethics, and behavioral science, she brings a human-centered lens to applied AI, treating quality not just as technical correctness but as trustworthiness, interpretability, and usefulness to the people who rely on it.

## Education

### Pennsylvania State University

PHD IN INFORMATICS, UNDER THE ADVISEMENT OF [DR. MARY BETH ROSSON](#)

University Park, PA.

2019 - 2024

### Shiv Nadar University

B.TECH. IN COMPUTER SCIENCE AND ENGINEERING

Greater Noida, UP.

2015 - 2019

## Experience

### Microsoft

SENIOR APPLIED SCIENTIST, MANAGER: JEREMY REYNOLDS

Redmond, WA.

Feb 2026 - Present

- Lead evaluation of the Finance Agent, designing LLM-as-a-judge frameworks to assess agent quality, accuracy, and reliability at scale.
- Drive systematically tuning prompts, instructions, and grounding data—to improve agent task performance on key quality metrics.
- Conduct root-cause analysis of agent successes and failure modes, translating findings into prioritized model, prompt, and product improvements.
- Designed scientifically grounded synthetic datasets to stress-test agent behavior across realistic and edge-case finance scenarios.

### Microsoft

AI PRODUCTIVITY SCHOLAR, ADVISOR: SUJAY JAUHAR

Redmond, WA.

Dec 2025 - Feb 2026

- Led a cross-functional initiative to evaluate the effectiveness and appropriateness of a starter prompt strategy.
- Built an LLM-as-a-judge evaluation framework, including synthetic dataset generation and quantitative metric design for scalable assessment.
- Integrated human-in-the-loop evaluation to validate and calibrate automated judgments.
- Partnered with Data Science and Product teams to align evaluation criteria with product goals and support deployment decisions.

### Artificial Intelligence and Human-Centered Computing, UT Austin

BULLARD RESEARCH FELLOW, ADVISOR: MATTHEW LEASE

Austin, TX.

Sept 2024 - Dec 2025

- Leading a user study on LLM usage in research workflows; identified patterns in tool adoption.
- Evaluating how different confidence representations affect user trust and calibration in AI-generated summaries.
- Designed AI-assisted annotation interface for low-resource language tasks aimed at improving accuracy and reducing cost.
- Mentored project on LLMs for research ideation and automated scientific workflows.

### Collaboration and Innovation Lab, Penn State

RESEARCHER, ADVISOR: MARY BETH ROSSON

University Park, PA.

Aug 2019 - Aug 2024

- Conceived and executed **end-to-end research project for peer-based adaptive learning** from scoping phase, participant recruitment and designing study plans to method creation and analysis
- Designed test-bed AI driven system to test student engagement on the platform
- Lead user initiative to understand user requirements and inform product development strategies for a cross-functional team
- Developed the prototype using research insights from preliminary studies
- Evaluated design implications that originate from a geographically distributed community

### INCITE institute, Columbia University

SCHOLAR, [LOGIC\(s\) MAGAZINE](#)

New York City, NY

Jan 2024 - July 2024

- Collaborated with the Logic(s) editorial board as a graduate-level Computing and Engineering student.
- Engaged in **cross-disciplinary discussions to address technological** violence and explore possibilities for radical transformations.
- Integrated **theoretical frameworks into technical discussions**, contributing to an understanding of the societal implications of technology.
- Contributed to dismantling perceived gaps between technical and humanities perspectives, affirming the significance of interdisciplinary approaches in addressing contemporary issues.

## PennHCI Lab, University of Pennsylvania

RESEARCH ASSISTANT, ADVISOR: DANAË METAXA

Philadelphia, PA.

June 2023 - Jan 2024

- Performed **LLM auditing for bias and discrimination identification** in script generation using various prompting techniques
- Implemented data pipeline to collect over 17,000+ user labelled TikTok videos
- Performed thematic analysis, data cleaning, topic modelling, sentiment analysis on the data collected
- Statistically analysed implicit and explicit nature of political data on TikTok using the data collected

## Human Language Technologies Lab, Penn State and Tech Policy Lab, University of Washington

RESEARCH ASSISTANT, ADVISOR: SHOMIR WILSON, AYLIN CALISKAN

University Park, PA.

July 2022 - July 2024

- Designed studies focused on ethical AI practices, **investigating biases in generative large language models** (LLM) such as GPT-4, Llama
- Developed methods to identify and mitigate nationality bias using zero-shot, few-shot, chain-of-thought prompting and instruction tuning
- Created **novel AI harms framework identifying cultural harms** within language and vision GenAI models
- Designed and thematically analyzed focus group and annotation study with Global South population to identify perceived bias and harms
- Evaluated quantitatively and qualitatively various LLM cultural bias alignment approaches

## Crowd AI Lab, Penn State

RESEARCH ASSISTANT, ADVISOR: KENNETH HUANG

University Park, PA.

May 2023 - July 2024

- Investigated an AI tool to support creative writing
- Analysed user interviews to capture user feedback on creative writing assistive tool
- Performed thematic analysis on the interview data and extracted design insights

## Supporting Civic Awareness Networks (SCAN) Lab, Penn State

RESEARCH ASSISTANT, ADVISOR: ANDREA TAPIA

University Park, PA.

May 2020 - Dec 2021

- Conducted and analysed user research on crisis response
- Performed user testing on situational awareness tool using interviews, surveys and usability studies
- Collaborated with cross-functional teams to identify research objectives
- Statistically analysed 3,100,000+ tweets to classify fake news and real news

## Dataminr

RESEARCH INTERN, HUMAN AI INNOVATION

New York City, NY (Remote)

Sept 2022 - Dec 2022

- Designed and executed three studies for human-in-the-loop AI system design: **Exploratory card sorting, dairy study and focus group study**
- Developed comprehensive needs assessment reports to inform key product development decisions, after collaboration with designers
- Communicated insights and findings with stakeholders, aligning with strategic business objectives

## Dataminr

RESEARCH INTERN, AI ENGINEERING

New York City, NY (Remote)

Sept 2021 - Dec 2021

- Designed, executed, and analyzed semi-structured interviews with human-in-the-loop experts
- Designed survey, collected data and analyzed results improving pipeline performance by 11%

## Publications

### 1. Conference

- (a) Kambhatla, G., **Gautam, S.**, Zhang, A., Liu, A., Srinivasan, R., Li, J. J., & Lease, M. (2025). **Improving the Distributional Alignment of LLMs using Supervision**. Accepted at Association for Computational Linguistics (**ACL 2026**)
- (b) **Gautam, S.**, Liu, H., Choi, Y., & Lease, M. (2026). **How Researchers Navigate Accountability, Transparency, and Trust When Using AI Tools in Early-Stage Research: A Think-Aloud Study**. Accepted at ACM Conference on Fairness, Accountability, and Transparency (**FAccT 2026**)
- (c) Ghosh, S., Venkit, P. N., **Gautam, S.**, & Ghosh, A. **What if AI systems weren't chatbots?** Accepted at ACM Conference on Fairness, Accountability, and Transparency (**FAccT 2026**)
- (d) Greenfield, J., Wang, S. T., Jung, S., **Gautam, S.**, Yang, K., & Metaxa, D. (2026, May). **What's Political on TikTok? Perceptions, Prevalence, and Patterns of Exposure to TikTok Users Perceive as Political**. In Proceedings of the International AAAI Conference on Web and Social Media (Vol. 20, No. 1, pp. 939-956).
- (e) Chakravorti, T., **Gautam, S.**, & Rajtmajer, S. (2025). **Open science practices among early-career human-computer interaction researchers in the US**. PloS one, 20(11), e0334692.
- (f) **Gautam, S.**, Chandra, M., De, A., Chakravorti, T., Malik, G., & De Choudhury, M. (2025, October). **Towards Experience-Centered AI: A Framework for Integrating Lived Experience in Design and Development**. In Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (Vol. 8, No. 2, pp. 1062-1077).

- (g) Ghosh, S., **Gautam, S.**, Venkit, P. N., & Ghosh, A. (2025, October). [Documenting Patterns of Exoticism of Marginalized Populations within Text-to-Image Generators](#). In Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (Vol. 8, No. 2, pp. 1107-1119).
- (h) **Gautam, S.\***, Ghosh, S.\*, Venkit, P. N.\*, Wilson, S., & Caliskan, A. (2024, October). [Do generative AI models output harm while representing non-Western cultures: Evidence from a community-centered approach](#). In Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (Vol. 7, No. 1, pp. 476-489).
- (i) Mahomed, Y., Crawford, C. M., **Gautam, S.**, Friedler, S. A., & Metaxa, D. (2024, June). [Auditing GPT's content moderation guardrails: can ChatGPT write your favorite TV show?](#). In Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (pp. 660-686).
- (j) Venkit, P. N.\*, Srinath, M.\*, **Gautam, S.**, Venkatraman, S., Gupta, V., Passonneau, R. J., & Wilson, S. (2023, December). [The Sentiment Problem: A critical survey towards deconstructing sentiment analysis](#). In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (pp. 13743-13763).
- (k) **Gautam, S.\***, Narayanan Venkit, P.\*, Panchanadikar, R.\*, Huang, T. H., and Wilson, S. [Unmasking Nationality Bias: A Study of Human Perception of Nationalities in AI-Generated Articles](#). AAAI/ACM Conference on AI, Ethics, and Society (**AIES 2023**).
- (l) Venkit, P. N., **Gautam, S.**, Panchanadikar, R., Huang, T. and Wilson, S. [Nationality Bias in Text Generation](#). European Chapter of the Association for Computational Linguistics (**EACL 2023**).
- (m) **Gautam, S.**, Akgun, M., and Mitra, P. [Exploring the challenges of AI experts to inform AI curriculum](#). ACM Technical Symposium on Computer Science Education (SIGCSE 2022).
- (n) Romera, G., **Gautam, S.** and Tapia, A. [Understanding Twitter's behavior during the pandemic: Fake News and Fear](#). Information Systems for Crisis Response and Management (**ISCRAM 2021**)
- (o) **Gautam, S.** and Rosson, M.B. [Exploring Feelings of Student Community across a Geographically Distributed University](#). Conference on Computer-Supported Cooperative Work and Social Computing (**CSCW, 2020**)

## 2. Journal

- (a) Karmelita, C., Carruthers, B., **Gautam, S.**, & Ferrara, A. [Outcomes of faculty training aimed at improving how allegations of research misconduct are handled](#). **Accountability in Research, 2025**, 1-30.
- (b) Grace, R., **Gautam, S.** and Tapia, A. [Continuity planning for public health crises: Designing workplace redundancies for organizational resilience](#). Journal of Emergency Management (**JEM 2024**)
- (c) Saurabh, S. and **Gautam, S.** [Modelling and statistical analysis of YouTube's educational videos: A channel Owner's perspective](#). Computers & Education 2019

## 3. Posters and Extended Abstracts

- (a) **Gautam, S.** [Co-designing AI for Scientific Discovery](#). **Cosmic Horizons Conference, 2025**.
- (b) **Gautam, S.** [Ethical AI in Citizen Science: Exploring Human-Agent Collaboration for Fact-checking](#). Grand Challenges for the Convergence of Computational and Citizen Science Research, 2025.
- (c) Huang, C. Y., **Gautam, S.**, Brooks, S. M., Lin, Y. F. and Huang, T. H. K. [Inspo: Writing Stories with a Flock of AIs and Humans](#). (**CHI 2025**)
- (d) **Gautam, S.**, Zhang, X., and Rosson, M.B. [Towards Dynamic Learning: A Framework for Simulating Adaptive Learning Systems](#). Companion of Computer-Supported Cooperative Work and Social Computing (**CSCW 2024**)

- (e) **Gautam, S.**, Rosson, M. B., & Akgun, M. [Exploring Potential Contributions of Social Learning to Adaptive Learning Systems](#).  
In Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems. (**CHI 2023**)

#### 4. Workshop

- (a) Lakshminarayanan, R. and **Gautam, S.** [Augmenting Collaborative Problem-Solving Through the use of GenAI: Opportunities and Challenges](#).  
Workshop on Augmenting Collaborative Problem-Solving: Exploring the Design and Use of GenAI for Groupwork **CSCW 2025**
- (b) **Gautam, S.** and Srinath, M. [Blind Spots and Biases: Exploring the Role of Annotator Cognitive Biases in NLP](#)  
Workshop on Bridging Human-Computer Interaction and Natural Language Processing (**NAACL 2024**)
- (c) **Gautam, S.** and Lakshminarayanan, R. [Balancing Act: Improving Privacy in AI through Explainability](#).  
Workshop on Human-Centered Explainable AI (**CHI 2024**)
- (d) **Gautam, S.**, Venkit, P., and Ghosh, S. [From Melting Pots to Misrepresentations: Exploring Harms in Generative AI](#).  
Generative AI and HCI Workshop (**CHI 2024**)
- (e) **Gautam, S.** What the Annotator Does Not Know, Can Harm Us.  
Workshop on Understanding and Mitigating Cognitive Biases in Human-AI Collaboration (**CSCW 2023**)

## Invited Talks

### Quant UX Conference, 2025

SPEAKER SERIES

Beyond the Patterns: Reclaiming the Human in UX Research with AI as a Thought Partner

USA

November 2025

### Grace Hopper Conference, 2025

SPEAKER SERIES

Designing the Future: Building Inclusive AI Through Human-Centered Thinking

Chicago, USA

November 2025

### Microsoft Research

HCI COMMUNITY TALKS

From Melting Pot to Misrepresentations: Understanding Human Perceptions of Cultural Bias in AI Generated Content

UK

October 2024

### University of Texas at Austin

ISCHOOL POSTDOC RESEARCH EXPO

The Learning Code: Designing AI-Driven Adaptive Learning Systems for Social Learning

Austin, USA

October 2024

### Pennsylvania State University

DATA STUDIES GROUP

Social Learning Using AI

State College, USA

March 2024

## Honors & Awards

|      |                                                                    |                       |
|------|--------------------------------------------------------------------|-----------------------|
| 2024 | <b>Liberatory Tech Fellow</b> , Logic(s) Magazine                  | Penn State            |
| 2023 | <b>Travel Grant</b> , Dept. of Information Sciences and Technology | Penn State            |
| 2023 | <b>Semi-Finalist (ChronoNews)</b> , Nittany AI Challenge           | Penn State            |
| 2023 | <b>Semi-Finalist (ClassCollab)</b> , Nittany AI Challenge          | Penn State            |
| 2020 | <b>Graduate Fellow</b> , Robert Graham Endowment                   | Penn State            |
| 2019 | <b>Graduate Fellow</b> , Robert Graham Endowment                   | Penn State            |
| 2015 | <b>Tuition Waiver</b> , Undergraduate Scholar                      | Shiv Nadar University |
| 2013 | <b>National Talent Search Scholar</b> , NCERT                      | Government of India   |

## Academic Service

### SCHOLARLY SERVICE

## Associate Chair

University of Texas at Austin &  
Pennsylvania State University

ACM/AAAI

2020-Present

- **Program Committeee** at AIES 2025
- **Associate Chair** for CSCW 2024 Posters, CSCW 2025 Full Papers, CSCW 2025 Posters
- Reviewed papers for various conferences and journals including CHI, CSCW, ISCRAM, IUI, i-Conference and Elsevier.

## Mentor

AAAI/ACM

CONFERENCE ON AI, ETHICS AND SOCIETY

2024

- Provided mentorship and feedback to PhD students.
- Moderated research discussion among student track scholars.

## Student Volunteer

AAAI/ACM

CONFERENCES BY ACM/AAAI

2020 - 2024

- Student volunteer at CSCW 2020, CSCW 2021, CSCW 2022, UbiComp 2021, CHI 2023 and FAccT 2023

## PROFESSIONAL SERVICE

### Millennium Scholars Program

Pennsylvania State University

MEMBER OF THE SELECTION COMMITTEE

2024

- Evaluate candidates' qualifications, including academic achievements, leadership potential, and commitment to increasing diversity in STEM fields.
- Collaborate with fellow interviewers and program coordinators to select scholars who demonstrate exceptional promise and dedication to making a positive impact in their fields and communities.

### Graduate Student Organization College of Information Sciences and Technology

Pennsylvania State University

PRESIDENT

2020-2021

- Spearheaded the organization's activities, including organizing talks, social events, and hackathons.
- Led efforts to foster both social and intellectual peer-to-peer connections within the IST community.
- Acted as a liaison between faculty and students, bridging communication and collaboration.

### HackData- The Data Science Hackathon

Shiv Nadar University

CO-ORGANIZER

Feb 2018

- Successfully co-organized HackData at Shiv Nadar University, Greater Noida, orchestrating a pan-India participation of over 200 students.
- Established key sponsorships, showcasing adept networking skills.
- Designed and implemented the comprehensive hackathon program, ensuring a dynamic and engaging experience for participants.

### SNU ACM-W Chapter

Shiv Nadar University

CHAIR

Nov. 2017 - Nov. 2018

- (Association for Computing Machinery - Women) Organized tutorials, talks and workshops to promote the presence of women in technology, as part as the founding committee of the SNU ACM-W Chapter